

Tiva Rocco

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[Linkedin](#) | [Personal Portfolio](#)

EDUCATION

North Carolina State University, Raleigh, NC

Ph.D. in Computer Science

Research Interests: CS Education, Software Engineering

Aug 2025 – Expected May 2030

California State University, Monterey Bay, Seaside, CA

B.S. in Computer Science — Software Engineering

Aug 2021 – Dec 2024

PUBLICATIONS & PRESENTATIONS

- [1] **Tiva Rocco** and Joshua B. Gross. 2024. Debugging Server- and Client-Side JavaScript. *Journal of Computing Sciences in Colleges (CCSC-NW)*, 40(2), 18-20. [*Conference Tutorial*]
- [2] **Tiva Rocco**, Md. Rawnak Saif Adib, Sarah Heckman, Jeffrey Carver, and Mark Sherriff. 2026. A Systematic Mapping Study on Teaming in Computer Science Education. *ACM Transactions on Computing Education (TOCE)*. [*Under Review*]

PROFESSIONAL DEVELOPMENT

University of Michigan, Ann Arbor, MI

NextProf Pathfinder — Participant

Sep 2026

Carnegie Mellon University, Pittsburgh, PA

LearnLab Summer School, Educational Data Mining Track — Certificate of Completion

July 2026

RESEARCH EXPERIENCE

NC State University, Raleigh, NC

Graduate Research Assistant, WISER Lab (Advisor: Dr. Sarah Heckman)

- Conduct CS education & software engineering research

Jan 2026 – Present

Blackwell Research Institute, New York, NY (Remote)

STEM Research Mentor

March 2026 – August 2026

- Guide high-school level students on reading scientific literature, forming research questions, writing a literature review, and submitting for publication

Under my mentorship:

Akzhanov, E. (2026). Strategic independence: Applying game theory to oligopolistic retail pricing. *The Blackwell Research Journal*, Edition VI, 46–47.

Desai, J. (2026). Artificial intelligence and stock market prediction: Effectiveness, limitations, and model comparison. *The Blackwell Research Journal*, Edition VI, 19–22.

Janiczek, K. (2026). Improving Formula 1 performance through calculus-based minimum lap-time optimization. *The Blackwell Research Journal*, Edition VII, [Accepted, Oct 2026]

Kuffel, J. (2026). Trends in unmanned autonomous systems' software and performance capabilities to influence future design. *The Blackwell Research Journal*, Edition VI, 49–50.

Liu, D. (2026). Artificial Intelligence in Diagnostic Radiology: Applications, Limitations, and Future Directions. *The Blackwell Research Journal*, Edition VII, [Accepted, Oct 2026]

California State University, Monterey Bay, Seaside, CA

Undergraduate Research Assistant, UROC (Advisor: Dr. Joshua Gross)

May 2024 – Dec 2024

- Developed software to engage student learning on computational complexity through visualization of energy consumption and environmental impacts of code
- Served as codebase manager offering debugging support to research team and overseeing workflows

TEACHING EXPERIENCE

CSC 326: Software Engineering, NC State University, Raleigh, NC

Fall 2025, Spring 2026, Fall 2026

Lead Graduate Teaching Assistant

- Lead lab instruction for 60+ students, grade assignments, hold office hours to enhance student development and periodically deliver full lectures to 200+ students

NC State Science House, Catalyst Program, Raleigh, NC

July 2026

AI & Organizational Skills Instructor

- Developed and taught two sessions on using AI for organizational skills, time management, and assignments to 60 STEM high school students with disabilities

Watsonville Tutoring Club, Watsonville, CA

Oct 2023 – June 2025

K-12 Mathematics, Reading, & Writing Tutor

- Implemented curriculums tailored to students' level of knowledge and goals for ages 5-17

California State University, Monterey Bay, Seaside, CA

Jan 2024 – Dec 2024

Senior Coordinator; CSUMB TA++ Program

- Offered mentorship and support for new and existing computer science TAs
- Held 3-5 training sessions per semester on TA expectations, ethics, and technical skills

California State University, Monterey Bay, Seaside, CA

Fall 2023, Spring 2024, Fall 2024

Web Development Teaching Assistant

- Assisted professor in leading lab instruction, graded assignments & exams, held office hours

INVITED PANELS

Graduate School Application Workshop – Alumni Panelist California State University, Monterey Bay (Virtual) Dec. 2025

LEADERSHIP & SERVICE

ACM and Computing Success Center, North Carolina State University, Raleigh, NC

Nov 2025 – Present

Workshop Organizer & Instructor

- Organize and lead GitHub-based personal portfolio workshops, providing free technical and professional development support for computing students

NC State Helping Hands Project, Raleigh, NC

March 2026 – August 2026

Mobile App Developer

- Built low-cost prosthetic hands for children, contributed to the complete iOS/Android mobile application from the ground up using React Native and TypeScript
- Built real-time Bluetooth Low Energy communication between the application and an Arduino-powered prosthetic, enabling children ages 7-12 to monitor their device's sensor activity, signal strength, and connection health

NC State Women in Computer Science (WICS), Raleigh, NC

Sep 2025 – May 2026

Student Mentor

- Served three undergraduate and one graduate student, meeting 2+ times per month to discuss coursework, career planning, and strategies for success to support confidence in computing

SIGCSE Technical Symposium, St. Louis, MO

Feb 2026

Student Volunteer

Community Builders of Monterey Bay, Seaside, CA

Aug 2023 – Oct 2023

Web Development Student Volunteer

- Developed a new site aimed at increasing youth involvement within the community using JavaScript, HTML, and CSS

Life Lab, Santa Cruz, CA

Jan 2023 – May 2023

Web Development Student Volunteer

- Led modernization and updates of the lab's website during a hosting platform migration

CERTIFICATIONS & AWARDS

NC State Graduate School, Raleigh, NC

June 2026

Advanced Teaching Certificate

- Developed an 8-week web development course aligned to learning outcomes and objectives, equipped with a syllabus, formative/summative assessment plan, and an AI policy all aligned with evidence-based teaching practices

- Presented 2 micro-teaching demonstrations to cohort to further develop teaching and presentation skills

NC State ACM Chapter, Raleigh, NC
Outstanding Graduate TA Award

April 2026

NC State Graduate School, Raleigh, NC
Teaching Certificate

March 2026